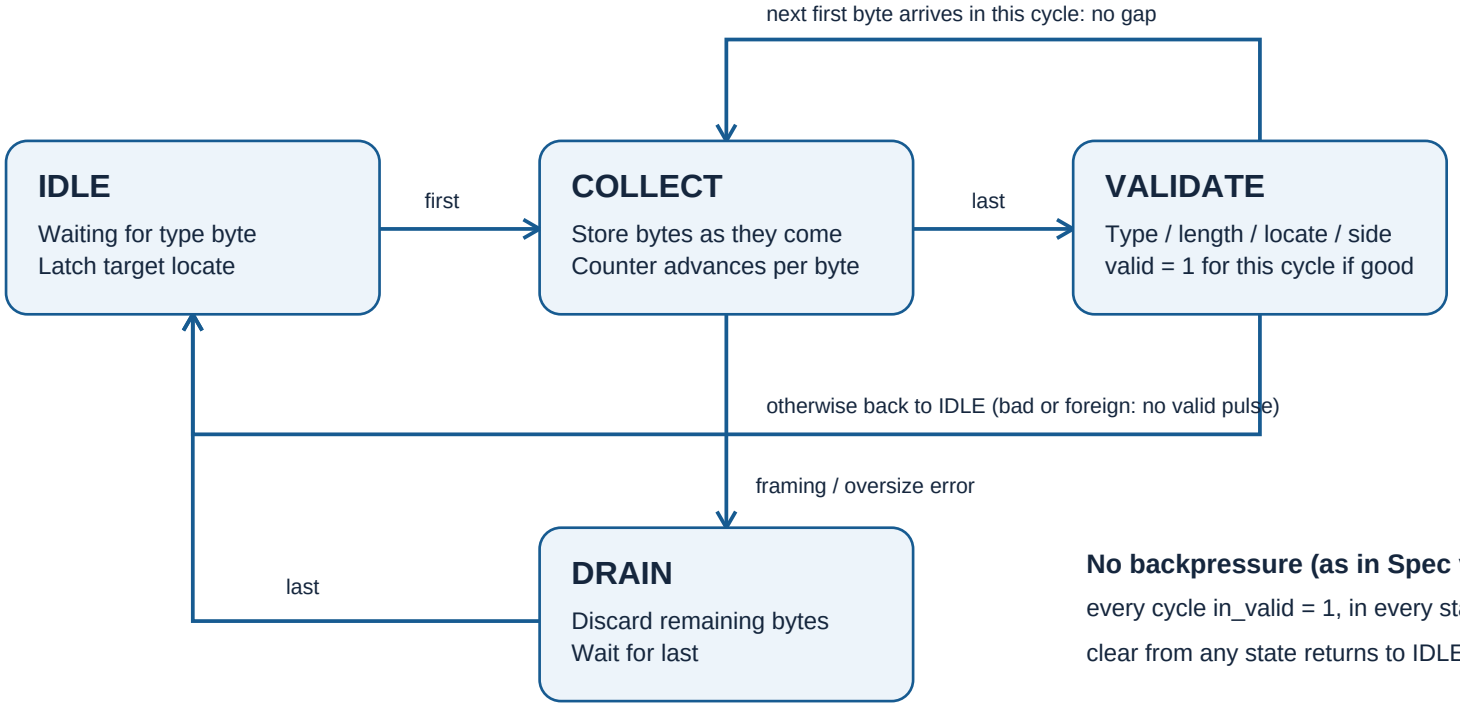
ITCH parser - state-machine sketch

Amirlan / CompE | November 8, 2026 | Implemented decoder

Input: one framed ITCH body, one byte per cycle while in_valid = 1. Outputs match Spec v1.



Outputs (Spec v1)

valid: one cycle per message
msg_type[1:0]: 0 A, 1 E, 2 X, 3 D
order_ref[63:0]: full wire ref
side: 0 buy, 1 sell (A only)
shares[31:0], price[31:0]

No backpressure (as in Spec v1): a byte is taken

every cycle in_valid = 1, in every state.

clear from any state returns to IDLE on the clock edge.

Boundary cases

First and last together: IDLE -> VALIDATE. Missing first: IDLE -> DRAIN (or IDLE if also last).

In COLLECT, a second first or a 37th byte rejects the frame. If that byte is also last, return directly to IDLE.

Unsupported type, wrong length or bad side never pulses valid. Foreign locate is filtered.

Worked Add Order

36 bytes: A / locate 1 / ref 42 / Buy / 100 shares / price 1,502,500 ($150.2500).

After the last byte at edge L: valid is high for one cycle; the next message may start in that cycle.

Scope: hw/itch_parser.ml. MoldUDP64 framing is a separate proposed adapter, not implemented here.

Full contract and transport FSM: docs/parser-interface.md. Tests: hw/test/test_itch_parser.ml.